

# **GFS SPECIFICATION**

## **SITE PREPARATION AND EARTHWORKS FOR ONSHORE LNG FACILITIES**

GFS IG.34.85.22-Gen.

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ECCN EAR99

### **GENERAL FUNCTIONAL STANDARD**



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## 1. INTRODUCTION

### 1.1 SCOPE

This new GFS specifies requirements for the design, planning, material selection, testing and construction and execution of onshore site preparation and earthworks.

This GFS sets out standard requirements based on general good practice for common site preparation works. This document shall need to be supplemented by a project-specific document for sites with difficult materials or conditions such as partially saturated soils, soils susceptible to collapse settlement, swelling soils, areas subjected to deep seasonal freeze/thaw cycles, peat, karst, highly erodible soils, gypsum and other dissolvable soils, highly saline soils, sabkha soils, carbonate soils, residual tropical soils, quick clays, liquefiable sands and silts, and areas where the soils and/or groundwater are highly contaminated with hydrocarbons or other problematic chemicals.

This GFS is classified as a TIER 2 document. See (1.3) for definitions of TIER 1/TIER 2.

### 1.2 DISTRIBUTION, INTENDED USE AND REGULATORY CONSIDERATIONS

Unless otherwise authorised by Shell GSI, the distribution of this GFS is confined to Shell companies and, where necessary, to Contractors nominated by them. Any authorised access to a GFS does not for that reason constitute an authorisation to any documents, data or information to which the GFS may refer.

This GFS is intended for use in LNG production facilities and associated supply/marketing. This GFS may also be applied in other similar facilities.

When GFSs are applied, a Management of Change (MOC) process shall be implemented; this is of particular importance when existing facilities are to be modified.

If national and/or local regulations exist in which some of the requirements could be more stringent than in this GFS, the Contractor shall determine by careful scrutiny which of the requirements are the more stringent and, which combination of requirements will be acceptable with regards to safety, environmental, economic and legal aspects. In all cases, the Contractor shall inform the Principal of any deviation from the requirements of this GFS which is considered to be necessary in order to comply with national and/or local regulations. The Principal may then negotiate with the Authorities concerned the objective being to obtain agreement to follow this GFS as closely as possible.

### 1.3 DEFINITIONS

#### 1.3.1 General Definitions

The **Contractor** is the party that carries out all or part of the design, engineering, procurement, construction, commissioning or management of a project or operation of a facility. The Principal may undertake all or part of the duties of the Contractor.

The **Manufacturer/Supplier** is the party that manufactures or supplies equipment and services to perform the duties specified by the Contractor.

The **Principal** is the party that initiates the project and ultimately pays for it. The Principal may also include an agent or consultant authorised to act for, and on behalf of, the Principal.

**TIER 1** packaged equipment solutions are kept evergreen and have a revision cycle.

**TIER 2** packaged equipment solutions do not have a revision cycle and updated when needed.

The word **shall** indicates a requirement.

If deviating from a requirement represented as **shall**<sup>1</sup>, refer to the Principal for the appropriate deviation process.

The word **should** indicates a recommendation.

The word **may** indicates a permitted option.

### 1.3.2 Specific Definitions

| Term                                 | Definition                                                                                                                                                                                                                                                                                                                    |
|--------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Cohesive materials</b>            | Soils classified per ASTM D2487 or ASTM D2488 as GC, SC, ML, CL, MH, CH, or materials classified as GM or SM when their fine fraction (material passing a No. 40 sieve) has a plasticity index of 4 or greater.                                                                                                               |
| <b>Excavation</b>                    | The removal, hauling, stockpiling and disposal of materials encountered within the specified excavation limits, regardless of the nature of the materials.                                                                                                                                                                    |
| <b>Fill</b>                          | The reclamation of fill material from a stockpile or borrow source that is transported, spread and compacted to form finished surfaces for intended use. Fill also includes backfilling of areas excavated of unsatisfactory material and the construction of embankments, berms and general fill.                            |
| <b>Granular materials</b>            | Soils classified per ASTM D2487 or ASTM D2488 as GW, GP, SW, SP, and materials classified as GM or SM when their fine function (material passing a No. 40 sieve) is non-plastic or has a plasticity index less than 4.                                                                                                        |
| <b>Inspection and Testing Agency</b> | The company, partner, or corporation retained to perform the inspections and tests required to determine and verify compliance with the work with the requirements of the project specifications.                                                                                                                             |
| <b>Modified Proctor density</b>      | The maximum dry density achieved per ASTM D1557 when testing a sample of materials representative of that to be compacted in the field.                                                                                                                                                                                       |
| <b>Optimum moisture content</b>      | The moisture content at which the soil maximum dry density is achieved in either the standard Proctor compaction test (ASTM D698) or the modified Proctor compaction test (ASTM D1557)                                                                                                                                        |
| <b>Relative Density</b>              | The degree of compactness of a free draining granular soil with respect to the loosest and densest conditions of the soil as determined by ASTM D4253 and ASTM D4254.                                                                                                                                                         |
| <b>Standard Proctor density</b>      | The maximum dry density achieved per ASTM D698 when testing a sample of materials representative of that to be compacted in the field.                                                                                                                                                                                        |
| <b>Topsoil</b>                       | The surface layer of soil with a high content of well-decomposed organic matter, suitable for vegetation or plants growth. Topsoil does not contain more than 5% (by volume) of stones/gravel, and it does not contain appreciable quantities of hard clods, stiff clay, brush, large roots, and other deleterious materials. |

**1.3.3 Abbreviations**

| <b>Term</b>  | <b>Definition</b>                          |
|--------------|--------------------------------------------|
| <b>ACI</b>   | American Concrete Institute                |
| <b>AFC</b>   | Approved for Construction                  |
| <b>ASTM</b>  | American Society for Testing and Materials |
| <b>CLSM</b>  | Controlled Low Strength Material           |
| <b>DEP</b>   | Design and Engineering Practice            |
| <b>GFS</b>   | General Functional Standard                |
| <b>HAZOP</b> | Hazard and Operability                     |
| <b>HAZID</b> | Hazard Identification                      |
| <b>HEMP</b>  | Hazards and Effects Management Process     |
| <b>HSE</b>   | Health, Safety and Environment             |
| <b>ITP</b>   | Inspection and Test Plan                   |
| <b>JHA</b>   | Job Hazard Assessment                      |
| <b>MCV</b>   | Moisture Condition Value                   |
| <b>MDD</b>   | Maximum Dry Density                        |
| <b>OMC</b>   | Optimum Moisture Content                   |
| <b>PPE</b>   | Personal Protective Equipment              |
| <b>PTW</b>   | Permit to Work                             |
| <b>PQP</b>   | Project Quality Plan                       |
| <b>QA/QC</b> | Quality Assurance/Quality Control          |
| <b>QCR</b>   | Quality Control Records                    |
| <b>QMS</b>   | Quality Management System                  |
| <b>SI</b>    | International System                       |
| <b>SOW</b>   | Scope of Work                              |
| <b>TBM</b>   | Temporary Bench Mark                       |

**1.4 CROSS-REFERENCES**

Where cross-references to other parts of this GFS are made, the referenced section or clause number is shown in brackets ( ). Other documents referenced by this GFS are listed in (7).

## 1.5 SUMMARY OF MAIN CHANGES

This is a new GFS.

## 1.6 COMMENTS ON THIS GFS

Comments on this GFS may be submitted to the Administrator using the same methods as for submitting comments on DEPs:

|                                                                      |                                                                                                                                                                                                                                                                   |
|----------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Shell DEPs Online</b><br>(Users with access to Shell DEPs Online) | Enter the Shell DEPs Online system at <a href="https://www.shelldeps.com">https://www.shelldeps.com</a><br><br>Select a GFS and then go to the details screen for that GFS.<br><br>Click on the "Give feedback" link, fill in the online form and submit.         |
| <b>DEP Feedback System</b><br>(Users with access to Shell Wide Web)  | Enter comments directly in the DEP Feedback System which is accessible from the Technical Standards Portal <a href="http://www.shell.com/standards">http://www.shell.com/standards</a> .<br><br>Select "Submit DEP Feedback", fill in the online form and submit. |
| <b>DEP Standard Form</b><br>(other users)                            | Use Standard Form PES 00.00.05.82-Gen. to record feedback and email the form to the Administrator at <a href="mailto:standards@shell.com">standards@shell.com</a> .                                                                                               |

Feedback that has been registered in the Feedback System by using one of the above options will be reviewed by the GFS Custodian for potential improvements to the GFS.

## 1.7 DUAL UNITS

This GFS contains both the International System (SI) units, as well as the corresponding US Customary (USC) units, which are given following the SI units in brackets. When agreed by the Principal, the indicated USC values/units may be used.

## 1.8 NON NORMATIVE TEXT (COMMENTARY)

Text shown in italic style in this GFS indicates text that is non-normative and is provided as explanation or background information only.

*Non-normative text is normally indented slightly to the right of the relevant GFS clause.*

## 1.9 UPDATE POLICY

This GFS will be updated when applied in a project in accordance with the framework for TIER 2 Packaged Equipment Solutions.

## 2. PROJECT REFERENCE DOCUMENTATION

### 2.1 APPLICABLE CODES AND STANDARDS

*This GFS is based on American standards and this is the minimum standard for site preparation and earthworks.*

### 2.2 PROJECT REFERENCE DOCUMENTS

1. The following documents shall form the project reference documents:
  - a. Geotechnical Factual Report;
  - b. Subsurface Detection Survey and relevant drawings of existing facilities;
  - c. Topographic Survey;
  - d. details of existing benchmarks;
  - e. HSE requirements;
  - f. temporary facilities and target drawings;
  - g. Environmental Impact Assessment and associated Environmental Survey Reports including soil and groundwater contamination report;
  - h. Flood Hazard Analysis Report.

### 2.3 EXISTING FACILITIES AND UNDERGROUND SERVICES

1. Prior to commencement of any site preparation works, reference shall be made to the Report of Subsurface Detection Survey and relevant drawings.

*Reference to the Report of Subsurface Detection Survey and relevant drawings ensures that no existing underground services are present that might be damaged or hinder the execution of the works under this specification.*

### 2.4 BENCHMARKS

1. The Contractor shall establish the requirement for new benchmark monuments for the project necessary for proper setting out for the work.
  - a. These benchmarks shall be updated as "As-built" condition in the drawings.
2. The Contractor shall plan the location of temporary/permanent site benchmarks taking into account future construction activities, the location of temporary facilities, materials and storage.
3. The location of site benchmarks shall allow for clear sight between site benchmarks.
4. All survey work shall be performed by qualified surveyors and under the direct supervision of a locally licensed or certified surveyor.
5. The Contractor shall plan the maintenance interval for all monuments and Temporary Benchmarks (TBMs).
  - a. The Contractor is responsible for maintenance of installed monuments and TBMs until completion of work.
6. Replacement and remediation procedures shall be provided by Contractor for any TBMs or site monuments found unstable.



**3. MATERIALS****3.1 SUITABLE FILL MATERIALS****3.1.1 General**

1. Unless otherwise specified, fill materials shall be from either on-site excavation, designated borrow areas or off-site sources.
2. The Contractor shall detail the source and provide specifications of any imported fill or materials used within the works.
3. Suitable fill materials shall be non-heaving, free of unsuitable materials (e.g., frozen lumps, organic matter, refuse, large stones or rocks, chunks of highly plastic clay, snow, ice, contamination) which might prevent proper compaction, or cause insufficient stability or bearing capacity for the superimposed loads to which it is designed to be subjected.

**3.1.2 Testing**

1. Tests of gradation, plasticity, density, and moisture content shall be performed for each type of fill material with a minimum of one set of tests every 2000 m<sup>3</sup> (2616 yd<sup>3</sup>).
2. The tests of gradation, plasticity, density, and moisture content shall be carried out in accordance with the standards specified in Table 3.1.

**Table 3.1 Reference Standards**

| <b>Test Method (all fill)</b>               | <b>Method</b>        |
|---------------------------------------------|----------------------|
| Classification                              | ASTM D2487           |
| Moisture content                            | ASTM D2216           |
| Standard sieve analysis                     | ASTM C136, D6913     |
| Liquid limit and plastic limit              | ASTM D4318           |
| Standard Proctor test/Modified Proctor test | ASTM D698/ASTM D1557 |

3. If the type or the source of fill material changes, the control tests shall be performed again using the new material.
4. Granular soils which exhibit no well-defined moisture-density relationship, test for index density, unit weight and relative density shall be in accordance with ASTM D4253 and ASTM D4254.
5. Sampling and testing of material at the source shall not preclude future rejection if it is subsequently determined to lack uniformity or it fails to conform to specified requirements.

**3.2 GENERAL FILL MATERIAL**

1. General fill materials shall meet following requirements:
  - a. liquid limit: 50 % (maximum);
  - b. plasticity index: 40 % (maximum);
  - c. maximum particle size: 2/3 thickness of the loose lift.

**3.2.1 General Cohesive Fill Material**

1. General cohesive fill material shall not be used as structural fill.
2. General cohesive fill shall comply with the properties specified in Table 3.2.

**Table 3.2 Grading requirements for general cohesive fill**

| Sieve size              | Percentage by mass passing the sieve |               |
|-------------------------|--------------------------------------|---------------|
|                         | Minimum value                        | Maximum value |
| 125 mm (5 in)           | 100                                  | -             |
| 2 mm (#10 sieve)        | 80                                   | 100           |
| 63 microns (#230 sieve) | 15                                   | 100           |

**3.2.2 General Granular Fill Material**

1. General granular fill shall comply with the properties specified in Table 3.3.
2. General granular fill material shall not be used as structural fill unless the result of embankment compaction trials can demonstrate acceptable properties (refer (4.5)) and meet with all the requirements stated in Table 3.5 (except grading).
3. If the Contractor proposes general granular fill for use as structural fill, the Contractor shall demonstrate the material suitability through compaction field embankment trials, in line with (4.5) of this GFS.
  - a. The embankment trial shall be at least 1 m height with plate load tests to demonstrate the satisfactory bearing capacity and permeability characteristics.
  - b. The Contractor shall provide detailed method statements for approval prior to any trial(s).
  - c. Contractor shall furnish a detailed geotechnical report providing geotechnical recommendations.

**Table 3.3 Grading requirements for general granular fill**

| Sieve size              | Percentage by mass passing the sieve |               |
|-------------------------|--------------------------------------|---------------|
|                         | Minimum value                        | Maximum value |
| 125 mm (5 in)           | 100                                  | -             |
| 63 microns (#230 sieve) | -                                    | 15            |

**3.2.3 Coarse Granular Fill Material**

1. Coarse granular fill shall comply with the properties specified in Table 3.4.
2. Coarse granular fill material shall not be used as structural fill unless the result of embankment compaction trials can demonstrate acceptable properties (refer (4.5)) which meet the requirements stated in Table 3.5 (except grading).
3. If the Contractor proposes general granular fill for use as structural fill, the Contractor shall demonstrate the material suitability through compaction field embankment trials, in line with (4.5) of this GFS.
  - a. The embankment trial shall be at least 1 m height with plate load tests to demonstrate the satisfactory bearing capacity and permeability characteristics.
  - b. The Contractor shall provide detailed method statements for approval prior to any trial(s).
  - c. Contractor shall furnish a detailed geotechnical report providing geotechnical recommendations.

**Table 3.4 Grading requirements for coarse granular fill**

|                         | Percentage by mass passing the sieve |               |
|-------------------------|--------------------------------------|---------------|
| Sieve size              | Minimum value                        | Maximum value |
| 300 mm (12 in)          | 100                                  | -             |
| 125 mm (5 in)           | 95                                   | -             |
| 600 microns (#30 sieve) | -                                    | 25            |
| 63 microns (#230 sieve) | -                                    | 15            |

**3.3 STRUCTURAL FILL MATERIALS**

1. Structural fill shall be of a granular, durable, non-heaving soil type that is not susceptible to change in properties through prolonged exposure to water, or local climatic conditions.
2. Structural fill shall be used as sub-base material for concrete paved areas, road construction and the bearing layer underneath foundations.

*The structural fill layer thickness will vary for each location according to site conditions/project drawing specifications.*

*Structural fill can also be used in general fill areas.*

3. Structural fill shall comply with the properties specified in Table 3.5 and Table 3.6.

**Table 3.5 Tests on structural fill material**

| Property                                                                                                                                                                                                                                                                                                                 | Test method      | Acceptance criteria                              |                 |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|--------------------------------------------------|-----------------|
|                                                                                                                                                                                                                                                                                                                          |                  | Minimum                                          | Maximum         |
| Liquid limit                                                                                                                                                                                                                                                                                                             | ASTM D4318       |                                                  | 40%             |
| Plasticity index                                                                                                                                                                                                                                                                                                         | ASTM D4318       |                                                  | 20%             |
| Grading                                                                                                                                                                                                                                                                                                                  | ASTM D6913, C136 | Refer Table 3.6                                  | Refer Table 3.6 |
| Soundness                                                                                                                                                                                                                                                                                                                | ASTM C88         | 85% and above for 5 cycles using sodium sulphate |                 |
| Uniformity coefficient <sup>1</sup>                                                                                                                                                                                                                                                                                      | (See Note 1)     | 10                                               |                 |
| Los Angeles coefficient                                                                                                                                                                                                                                                                                                  | ASTM C535        |                                                  | 40              |
| Note: 1. Uniformity coefficient is defined as the ratio of the particle diameters $D_{60}$ to $D_{10}$ on the particle size distribution curve, where:<br><br>$D_{60}$ = particle diameter at which 60% of the soil by weight is finer;<br><br>$D_{10}$ = particle diameter at which 10% of the soil by weight is finer. |                  |                                                  |                 |

**Table 3.6 Grading requirements for structural fill**

|                         | Percentage by mass passing the sieve |               |
|-------------------------|--------------------------------------|---------------|
| Sieve size              | Minimum value                        | Maximum value |
| 75 mm (3 in)            | 100                                  | -             |
| 63 microns (#230 sieve) | -                                    | 15            |

### 3.4 BEDDING MATERIAL

1. Pipeline bedding material shall meet the gradation limits in accordance with ASTM D6913 as specified in Table 3.7.
  - a. For pipelines in remote areas where the cost of obtaining 'ideal' bedding material may be high, alternative gradings for pipeline bedding materials may be proposed for review and approval.

**Table 3.7 Tests on bedding material**

|                         | Percent passing |               |
|-------------------------|-----------------|---------------|
| Sieve size              | Minimum value   | Maximum value |
| 4.75 mm (#4 sieve)      | 100             | -             |
| 2 mm (#10 sieve)        | 80              | -             |
| 75 microns (#200 sieve) | -               | 10            |

## 3.5 CONTROLLED LOW-STRENGTH MATERIAL

1. Controlled Low-Strength Material (CLSM) may also be used as a replacement for structural fill.
2. CLSM or “flowable fill” shall conform to ACI 229.
3. The required value of compressive strength shall be determined by the project-specific application.
  - a. In the absence of project specific guidelines, the mix design shall be proportioned to produce a minimum 28-day unconfined compressive strength between 0.69 MPa and 1.38 MPa (100 psi and 200 psi).
4. The mix design shall include field trials using the actual raw materials.
  - a. Test and reporting requirements are given in Table 3.8.
5. All information shall be submitted for review prior to use/incorporation on the project.

**Table 3.8 CLSM Mix Design Trails**

| Material                               | Test Each Mix Design                                                    | Batch Plant    | Per Trail Trench |
|----------------------------------------|-------------------------------------------------------------------------|----------------|------------------|
| CLSM                                   | Cube/cylinder compressive strength 7 days, 14 days, 28 days and 56 days | 1 set          | As directed      |
|                                        | Slump flow                                                              | 1              | 1                |
|                                        | In-situ field density at 2 days, 7 days and 14 days                     | N/A            | 1                |
|                                        | Moisture content at 2 days, 7 days and 14 days                          | N/A            | 1                |
|                                        | Batch records                                                           | Each Batch     |                  |
|                                        | Visual inspection                                                       | Daily          | Daily            |
|                                        | Excavation Trials >56 days using hand tools and small excavator.        | N/A            | As directed      |
|                                        | In-situ thermal resistivity tests                                       | N/A            | As directed      |
| Aggregate                              | Particle size distribution, chemical and Proctor tests                  | Prior to Trial |                  |
| Cement, water, admixtures & mix design | Material/Chemical tests and certificates                                | Prior to Trial |                  |

5. Test and reporting requirements shall be as given in Table 3.9 and relevant ASTM standard and as directed.

**Table 3.9 CLSM Production QA/QC Testing**

| <b>Material</b>                        | <b>Test Each Mix Design</b>                                 | <b>Batch Plant</b> | <b>Per Trail Trench</b> |
|----------------------------------------|-------------------------------------------------------------|--------------------|-------------------------|
| CLSM                                   | Cube/cylinder compressive strength 7 days and 28 days       | 1 set              | As directed             |
|                                        | Slump flow Point of Delivery                                | 1                  | As directed             |
|                                        | In-situ field density at 2 days and 7 days                  | 2                  | As directed             |
|                                        | Moisture content at 2 days and 7 days                       | 2                  | As directed             |
|                                        | Batch Records                                               | Each Batch         |                         |
|                                        | Visual inspection                                           | Daily              | Daily                   |
|                                        | In-situ thermal resistivity tests                           | N/A                | As directed             |
| Aggregate                              | Particle size distribution, chemical and Proctor tests      | As directed        | As directed             |
| Cement, water, admixtures & mix design | Material/chemical tests and certificates                    | Monthly            |                         |
| QA/QC Monitoring and reporting         | Graphical representation of all QA/QC testing on date basis | Weekly             |                         |

### 3.6 RIP-RAP

1. Rip-Rap and Stone Pitching material shall be used for erosion control of culvert inverts, ditches, side slopes and riverbanks as shown on the drawings.
2. The rock used for Rip-Rap shall be hard and durable rock with individual particles being generally cubic in shape.
3. The rock used for Rip-Rap shall be free from all deleterious matter including clay and weathered seams and shall have a density  $>2.5 \text{ Mg/m}^3$ .
4. Specialised technical advice shall be sought to determine strength and durability requirements for rip rap that is to be used in high exposure conditions, such as for protection of slopes in coastal locations subject to wave action.

### 3.7 GEOTEXTILES AND GEOGRIDS

1. The geotextile or geogrid materials used for a specific application shall be based on the individual material having adequate strength, long-term strain characteristics, permeability, thickness, porosity and roughness properties necessary to satisfy the requirements for that application and the design life specified in the project specification.
2. Manufacturers product data including installation, overlapping and joining details, handling, storage and repair instructions shall be provided for review.
3. For geogrid and geotextile materials with bi-directional engineering properties, the intended plan orientation of the material shall be stated.

**3.8 WATER**

1. Water for site preparation work shall be clean and free of detrimental impurities such as oils, acids, alkali, organic matters, chlorides, sulphates and deleterious matter.
2. Where the source of water for site preparation is from a potable water source, testing of water quality is not required.
3. Where the requirements of (3.8, items 1 or 2) cannot be met, an approved materials testing laboratory shall perform water quality tests with the acceptance criteria in Table 3.8.

*In remote locations, where it is difficult to source 'clean' water, relaxation of the specified limits for chlorides, sulphates, pH and total solids, may be considered if the consequences for corrosion of buried concrete and steel are clearly understood and mitigated, including any determination of potential alteration of soil properties by appropriate testing. Contractor shall seek specialist technical advice and guidance should this approach be taken.*

**Table 3.8 Tests performed on water**

| Property     | Test Method | Acceptance Criteria (maximum limits) |
|--------------|-------------|--------------------------------------|
| Chlorides    | ASTM D512   | 200 mg/l (0.027 oz/gal)              |
| Sulphates    | ASTM D516   | 500 mg/l (0.067 oz/gal)              |
| pH           | ASTM G51    | 6.5 to 9.0                           |
| Total solids | ASTM C1603  | 4000 mg/l (0.534 oz/gal)             |

**4. EXECUTION****4.1 PROVISION OF TEMPORARY FACILITIES****4.1.1 General**

1. Requirements for temporary facilities shall be detailed in the Project Scope of Work (SOW).

**4.1.2 Temporary Access**

1. The Contractor shall lay and maintain all necessary temporary roads throughout the contract period to provide all-weather access for all construction activities.

*It is preferred that temporary roads can coincide with the planned permanent routes.*

2. Disturbance of permanent surfaces and subsurface drainage shall be minimised.

**4.1.3 Temporary Lay-down Areas**

1. Temporary lay-down areas shall be located within the designated Contractors area and include suitable access, temporary paving and surface drainage, as necessary.
2. All temporary lay-down areas required for construction activities shall be levelled to suit the planned lifting operations.
3. The bearing capacity of the temporary lay-down areas shall be determined such that lifting operations can be planned and carried out safely.
4. Access to the temporary lay-down areas for lifting equipment and construction vehicles shall be maintained.

**4.1.4 Erosion Control**

1. The Contractor shall provide for review an Erosion Control Plan specifying measures to control erosion by both wind and water run-off during and after construction.

**4.1.5 Drainage and Dewatering**

1. The Contractor shall submit a plan for temporary drainage prior to the start of any earthworks to ensure drainage of laydown areas, excavations, cuts, fills, stockpiles and borrow areas to prevent ponding of surface water.
2. Temporary drainage work shall comprise design and installation of open channel ditches, subsurface drains, culverts, tie-ins and the construction of miscellaneous structures as indicated on the project drawings.
3. Temporary drainage facilities shall be removed at the completion of site construction, or as specified in the Project SOW.
4. The Contractor shall be responsible for ensuring the quality of water discharged from the site boundaries are compliant with local and national regulations and include measures to monitor the water quality and, if necessary, contain and treat water before discharge.
5. Existing drainage ditches and watercourses shall be temporarily or permanently rerouted or backfilled as indicated in the Project SOW and executed according to the requirements set out in this GFS.
6. If the permanent surface water drainage system is to be used during construction, rather than a temporary network developed by the Contractor, then measures shall be taken to prevent siltation.
7. Prior to handover the Contractor shall be responsible for full cleaning to the base of ditches and carry out any associated reinstatement works.

**4.1.6 Fencing**

1. The Contractor shall be responsible for providing temporary fencing to barricade the areas where site preparation is carried out.
2. Fencing shall be of 2.0 m (6.5 ft) height and continuous to the ground unless otherwise specified in the Project SOW.

**4.2 SITE PREPARATION****4.2.1 Demolition**

1. Demolition shall consist of the removal of existing structures, facilities and abandoned pipelines to the extent marked on the drawings in the Project SOW.
2. The Contractor shall develop a detailed work procedure for the demolition of existing facilities where applicable, including measures to protect personnel and the adjacent facilities and buildings designated to remain after all construction works.
3. The Contractor shall take measures to protect those items to be preserved such as existing foundations, underground obstacles and adjacent buildings from damage.



#### 4.2.2 Clearing and Grubbing

1. Clearing and grubbing shall consist of the removal and disposal of vegetation, trees, stumps, roots, logs, rubbish and other material unfit for reuse except for those trees and shrubs marked for preservation.
2. The Contractor shall take measures to protect trees, shrubs and other landscape features marked for preservation in the Project SOW.
  - a. Trees, bushes or shrubs outside the construction area shall not be damaged.
3. In the areas designated for clearing and grubbing; trees, stumps and other vegetation shall be removed to the bottom of their root zone unless otherwise specified in the SOW.
4. Depressions made while clearing and grubbing operations shall be backfilled with material suitable for the end use of the ground and compacted to conform to the original ground level.

#### 4.2.3 Stripping

1. A topographic survey shall be done before and after stripping operations to establish an earthworks baseline.
2. Stripping shall consist of the excavation, removal and satisfactory disposal of all topsoil and soil containing organic material.
  - a. The excavation depth shall be to the bottom of the root zone but not less than 150 mm (6 in).
3. Stripped topsoil shall not be adulterated with other excavated soil.
4. The Contractor shall ensure stripping activities which could damage existing structures are carried out by hand-operated machines.

#### 4.2.4 Preparation of subgrade

1. After stripping is completed, and no more excavation is planned, the ground surface shall be prepared for fill placement or as a finished subgrade per the requirements of (4.2.3).
2. Unsuitable material in areas to receive fill shall be removed and replaced with suitable material.
3. Cut areas to receive fill or finished cut surfaces shall be proof rolled to check for elasticity or deformation under the weight of a vehicle weighing at least 25 t (27.5 ton).

*Normal practice is for this to be assessed by visual inspection of the ground during rolling to determine whether large-scale elasticity or deformation is visible.*
4. If the subgrade displays elasticity or deformation during the proof roll, the material shall be excavated and re-compacted, or removed and replaced with fill material meeting the requirements of this GFS.
5. In areas to receive fill; scarify, moisture condition and re-compact the upper 150 mm (6 in) of in-place materials as follows:
  - a. to at least 95 % of the maximum dry density per ASTM D1557 under roads, structural paving or below structural foundations;
  - b. to at least 90 % of the maximum dry density per ASTM D1557 in general areas.
6. In areas subject to (4.2.4, item 5), proof rolling shall be repeated in accordance with (4.5).
7. For area paving, a minimum of 300 mm (12 in) non-frost susceptible fill shall be placed if frost susceptible soils are present below area paving.

#### 4.3 EXCAVATION

##### 4.3.1 General

1. The Contractor shall remove soil, rock and other materials as necessary to achieve the required finished grades, slopes and subgrades.
2. Satisfactory materials resulting from the excavation shall be used for the construction of fills and embankments and for the replacement of removed unsatisfactory materials.
3. Where excavations exceed 1.2 m (4 ft) in depth, the excavation side walls shall be sloped, or shored to prevent cave-ins or other hazardous conditions.
4. The Contractor shall be responsible for maintaining all slopes and sides of excavations in a safe and stable condition until final acceptance of the work.
5. Precautions shall be taken to protect adjacent work and structures and to prevent detrimental effects to adjacent foundations.
6. Shoring adjacent to existing underground installations during construction shall be designed and detailed on the drawings based on a geotechnical report(s) by the Contractor's geotechnical engineer.
7. Excavation shall not be permitted under foundations.
8. Excavated reusable materials shall be transported to a stockpile area and stored as described in this GFS.
9. Materials that are not reused shall be disposed of in accordance with the Project SOW and applicable regulations.
10. Any over-excavation shall be backfilled to the required level with compacted structural fill, or concrete.
  - a. The details of over-excavation and backfilling shall be recorded on the relevant Quality Control Record (QCR).
11. To avoid excess surcharge at the top of excavations, no materials or heavy equipment shall be located near the edge of the excavation as defined by (4.3.1, item 12).
12. A Geotechnical Engineer shall review the required distance to the edge of the excavation for each location.
  - a. The exclusion area shall be marked by pegs.
13. The stockpiled material shall be kept a minimum of 900 mm (3 ft) from the edge of the excavation.
14. Excavated areas shall be barricaded along with the designated entry/exit points.

##### 4.3.2 Structural Excavation

1. Structural excavations shall include excavations for footings, pile caps, grade beams, pits, basements and retaining walls.
2. Structural excavations shall be performed according to the dimensions, grades and elevations indicated on the drawings.
3. If the excavated sub-grade, under foundation exposes hard, sound rock larger than 76.2 mm (3 in) and the field Geotechnical Engineer concurs that it would compromise performance, then the sub-grade shall be over-excavated to a minimum of 300 mm (12 in) and backfilled with compacted structural fill.
4. If the bottom of a structural foundation is above frost depth, non-frost susceptible fill shall be placed underneath the foundation down to frost depth.
5. The Contractor shall remove all unsuitable materials encountered at footing grades and replace those materials with compacted structural fill.

6. The Contractor shall protect the soil adjacent to and beneath the foundations and the bottom of the excavations from frost action.
7. The Project SOW shall detail specific measures for preparation of sub-grade materials for footing.
8. Excavations shall only be kept open for as long as required to perform the work.
9. Before backfilling around foundations, all formwork shall be removed in a safe manner and the excavation cleared of debris, waste and other deleterious materials.
10. Excavation and backfilling around existing foundations shall be carried out carefully in order not to damage the existing foundations.

*Access for heavier compaction equipment is often difficult around the foundation backfill, alternative smaller compaction plants (such as whacker plates coupled with thinner compaction layers) can be used where required to ensure adequate compaction around foundations.*

#### **4.3.3 Trenching**

1. Excavation of trenches for culverts and pipe shall be to the elevation of the bottom of the pipe/cable and to a width necessary to allow satisfactory working room.

#### **4.3.4 Trial Excavation for Brownfield Site**

1. The Contractor shall conduct trial excavations, required by the Project SOW, necessary for locating the position of underground services by hand or by hydro vac.
2. Services or other obstructions shall be physically marked on the ground when encountered in the course of excavation existing services or other obstructions.
  - a. As-built drawings shall be updated to reflect services or other obstructions.

### **4.4 STOCKPILING AND SEGREGATION OF MATERIALS**

#### **4.4.1 Segregation of Materials**

1. The Contractor shall segregate materials to meet the material quality and gradations as follows:
  - a. topsoil;
  - b. structural fill (foundations, concrete paved areas);
  - c. general fill (embankments, laydown areas);
  - d. natural aggregates (stones and rocks);
  - e. broken concrete or asphalt (or contaminated soil);
  - f. organic matter (plants and shrubs).
2. Excess or unsuitable materials shall be disposed of in accordance with the Project SOW.
3. Segregation and quantification of hazardous and non-hazardous waste material and the certified disposal of hazardous material shall be carried out in accordance with the Project Waste Management Plan.

**4.4.2 Stockpiling**

1. Stockpiling shall include the storage of materials required for re-use from other site preparation activities in a safe location that is protected from damage or other deterioration.
2. The Contractor shall ensure that stockpiling of materials will not change their properties and composition thus rendering them unfit for use.
3. Stockpiled topsoil or other grubbed material shall not be stockpiled on a prepared subgrade and shall be stored separately from all other materials.
4. The height of the stockpile shall not be greater than 5 m (16.4 ft) for general soils and 3 m (9.8 ft) for topsoil or the height according to the material to be stockpiled at a slope 1V:2H.
5. The Contractor shall arrange stockpiles so as not to obstruct drainage or other construction operations and shall have signs to identify material type/classification to be used in the works.
6. Stockpiles, stockpile areas, laydown areas and any other area which cause a dust hazard shall be sprayed with water or an approved binder, covered, or otherwise sealed in order to eliminate the dust hazard.
7. All areas utilised for stockpiling shall be restored after completion of all earthworks as per project requirements.

**4.5 COMPACTION TRIALS****4.5.1 General**

1. Where compaction plant is utilised for compaction of fill, the Contractor shall conduct successful compaction trials on the proposed fill materials using the proposed compaction plant.
2. The objective of the compaction trials shall be to demonstrate the proposed compaction plant and equipment achieves a dry density that meets the specified minimum dry density.
3. The Contractor shall perform the required laboratory tests to define the engineering properties of the proposed material as defined in (3).
  - a. The field trial shall commence once the required laboratory test results are available.
4. The compaction trial areas shall not be incorporated into permanent fill unless the work complies with the requirements of this GFS.

**4.5.2 Procedure**

1. Three test areas of 20 m (65 ft) length by 6 m (20 ft) width, or three roller widths, whichever is wider shall be prepared on the actual site of construction from which the topsoil has been removed and the formation level prepared as described in (4).
2. The fill material under trial shall be spread over each of these three areas.
  - a. The minimum thickness of the layer shall be 50% larger than the maximum particle size.
  - b. The depth of the loose material in the strips shall be at least 150 mm (6 in), 300 mm (12 in) and 450 mm (18 in) in each of the test areas respectively.
3. The moisture content of the proposed fill material shall be determined at three locations in each of the three trial areas to confirm it follows Table 4.1.

4. Rapid determination of moisture content may be made to avoid a delay in using the oven drying method.
  - a. Rapid determination of moisture content shall be supported by additional samples being taken and tested using the oven drying method.
5. If the fill material fails to comply, then the Contractor shall take actions to bring the moisture content within limits specified in Table 4.1.
6. The Contractor shall determine the elevation at six locations of both the top surface of the fill material and the surrounding formation level.
7. The Contractor shall propose the number of passes 'n' required to achieve the specified minimum dry density for the proposed fill material.
8. The test areas shall then be compacted with the plant using 'n-2' number of passes.
9. Samples shall be taken and tests performed at a minimum of five locations (a minimum of two from mid-height and two from the base of the layer) on each test area to determine the moisture content and dry density as well as the Moisture Condition Value (MCV) and any other properties of the material over the full thickness of the layer of fill material.
10. For each test area, the soil density shall be determined as the mean of five tests carried out using one of the following methods:
  - a. Drive cylinder method (ASTM D2937);
  - b. Nuclear methods (ASTM D6938);
  - c. Rubber balloon method (ASTM D2167);
  - d. Sand-cone method (ASTM D1556);
  - e. Sand replacement (ASTM D4914);
  - f. Water replacement (ASTM D5030).
11. If nuclear density gauge or gamma-radiation test methods are used to determine compliance of placed material other than that specified in this GFS, the test method shall be calibrated against the field density tests specified in this GFS.
12. Nuclear density and gamma radiation tests shall not replace traditional test methods until results of a successful calibration exercise have been submitted for review for each different soil type subject to field compaction.
  - a. This calibration exercise may be carried out during the compaction trial.
13. The three areas shall be compacted with another two passes, followed by another suite of sampling and testing as described in (4.5.2, item 10) at different locations to the previous points.
  - a. A final two passes shall be carried out followed by a final suite of sampling and testing as described in (4.5.2, item 10) at different points.
14. If the Contractor considers the compactive effort from the equipment is adversely affected by being carried out directly on the formation level, the Contractor may repeat the compaction trial on an additional layer placed on top of the first compacted layer.
15. Once the compaction of each layer is complete, the surface elevation of the fill shall be determined at the same locations as the pre-compaction survey.
16. All compaction equipment shall have a sign attached that states each type of fill, the layer thickness and the number of passes required to achieve 90 % and 95 % compaction respectively, whether this relates to standard Proctor compaction test (ASTM D698) or to modified Proctor compaction test (ASTM D1557, and whether the equipment is to be operated with or without vibration

17. The dry density and moisture content test results shall be compared against the maximum dry density that is achieved at the optimum moisture content to determine the percentage of compaction achieved in the construction trial.

- a. This degree of compaction shall be compared with the requirements of (4.7).

18. The trials shall be completed before the main construction begins.

#### **4.5.3 Compaction trials using alternative plant on approved fill material**

1. If the Contractor proposes to use alternative plant and equipment during construction, the adequacy of the alternative equipment shall be proven, and the demonstration witnessed and approved by the Contractors Geotechnical Engineer.
2. The demonstration shall include the following process as a minimum:
  - a. An area of 20 m (65 ft) length by 6 m (20 ft) wide, or three roller widths, whichever is wider shall be prepared on the actual site of construction from which the topsoil has been removed.
  - b. The trial may be performed on the formation level prepared, as described in this GFS, or on fill material placed using approved materials and equipment.
  - c. The fill material shall be spread over the prepared area in a thickness for which the equipment is to be tested, but shall not exceed 450 mm (1.5 ft).
  - d. The moisture content of the proposed fill material shall be determined at three locations in the trial areas to confirm compliance with the range specified in Table 4.1.
  - e. If the fill material fails to comply, then the Contractor shall take actions to bring the moisture content within limits specified in Table 4.1.
3. The Contractor shall propose the number of passes 'n' that are expected to be required to achieve the specified minimum dry density for the proposed fill material.
  - a. The test area shall then be compacted with the plant using 'n-2' number of passes.
  - b. Samples shall be taken and tests performed at a minimum of five locations (a minimum of two from mid-height and two from the base of the layer) on the test area to determine the as placed properties of the fill material over the full thickness of the layer as required in the approved method statement.
4. The area shall be compacted with another two passes, followed by another suite of sampling and testing at different locations to the previous points.
5. This process shall be repeated until the results satisfy the acceptance criteria for the fill and compaction equipment under test.

#### **4.6 FILL AND EMBANKMENTS**

##### **4.6.1 General**

1. Fill shall be placed to the lines and grades and cross sections as shown on design drawings.
2. Before placement of fill, the subgrade shall be prepared in accordance with (4.2.4).
3. Fill shall include materials excavated from the site or imported from outside that meet the requirements of this GFS and additional requirements in SOW for their intended use.
4. General fill may be used for filling low-lying areas to design grade where roads and foundations are not planned, and to the depths, as shown on the project-specific drawings.

5. Structural fill material shall be used below all foundations, slabs, roads and paving at the locations, and to the depths, as indicated on the project-specific drawings.
6. Fill materials shall be compacted to no less than the requirements given in Table 4.1.

*Different compaction criteria may be employed depending on the project-specific application and the local soil properties. For example, compaction criteria may be defined in relation to not exceeding a "maximum air voids" value. Different levels of compaction may be specified for granular soils and for clay soils since the latter can prove problematic if over-compacted dry of optimum, and it is often considered preferable to accept a lower level of compaction at a moisture content at or exceeding optimum in order to minimise the potential for future heave or settlement.*

**Table 4.1 Minimum compaction requirements of general fill areas**

| <b>Material classification</b> | <b>Modified Proctor density<br/>(ASTM D1557)</b> | <b>Standard Proctor density<br/>(ASTM D698)</b> | <b>Relative density<br/>(ASTM D4253,<br/>ASTM D4254)</b> |
|--------------------------------|--------------------------------------------------|-------------------------------------------------|----------------------------------------------------------|
| Structural fill & bedding      | 95 %                                             | 98 %                                            | 80 %                                                     |
| General fill                   | 90 %                                             | 95 %                                            | 70 %                                                     |

7. The following tests to measure in-place density and moisture content for compacted fill shall be performed in accordance with following standards unless specified otherwise in the Project SOW:
  - a. Drive-Cylinder Method (ASTM D2937);
  - b. Nuclear Method (ASTM D6938);
  - c. Rubber Balloon Method (ASTM D2167);
  - d. Sand-Cone Method (ASTM D1556);
  - e. Sand replacement (ASTM D4914);
  - f. Water replacement (ASTM D5030).

#### **4.6.2 Backfill**

1. Backfill shall not be placed around new cast-in-place concrete structures until the concrete has cured for at least three days or has achieved 60% of the specified strength.
2. The Contractor shall ensure that backfill is brought up simultaneously on all sides of a structure to avoid the development of unbalanced earth pressure loads on the structure.
3. Step excavation side slopes with each layer of backfill to avoid the development of unnecessary loads against the structure caused by backfill wedging between the structure and the excavation sidewalls.
4. The Contractor shall not disturb any adjacent structure by filling operation.
5. Fill or backfill around piping, conduits, cables and sewers shall be bedding material as per the drawings and Project SOW.
  - a. The backfill material shall be placed carefully by hand simultaneously on both sides of the line to prevent side loading in layers up to 150 mm (6 in) and thoroughly compacted until the line is covered with minimum 300 mm (12 in) of fill.

6. While backfilling trenches for cables and pipes, the following shall apply:
  - a. protective coating not damaged;
  - b. pipe joints left exposed until necessary tests have been performed.
7. Use of CLSM for backfill around foundations, haunches of pipes and other areas is permitted provided an approved mix design meets the requirements of (3.5) of this GFS.
8. If the Contractor causes damage as a result of the backfill operation, the Contractor shall re-excavate, carry out necessary repairs and re-compact material to the required specification.

#### 4.7 COMPACTION

1. Compaction shall begin only after placing, spreading and controlling the water content to the required dry densities and moisture content in accordance with the Project SOW.
2. Compaction equipment shall be capable of achieving the specified degree of compaction, as used in the compaction trial, and compact the fill material with the same number of passes as established in field compaction trials.
3. The Contractor shall test each layer for the required density before starting with next layer.
4. Each layer of fill shall be compacted to the requirements of Table 4.1.
5. The Contractor shall rework any layer or portions until the material meets the specified density.
  - a. Where the dry density requirements cannot be achieved, the material shall be removed and replaced with suitable fill material.
6. The moisture content of the material being compacted shall be within +/- 2% of the optimum moisture content as determined by the applicable test method unless specified otherwise in, project SOW.
  - a. The Contractor shall condition the moisture content of fill materials as necessary to achieve the required moisture content.
  - b. When the moisture content  $> \pm 3\%$ , defer compaction until the material has dried to a suitable moisture content.
  - c. Natural drying may be accelerated by manipulation to increase the rate of evaporation, or by blending in dry material, take care not to exceed the specified maximum layer thickness for compaction.
7. Where the width of fill to be compacted is wider than the width of the roller, the overlap of the adjacent trips shall not be less than 450 mm (18 in).
8. Fill material shall be placed in loose lifts not exceeding the thickness as given in Table 4.2.

**Table 4.2 Fill classification and thickness**

| Fill Classification | Maximum Thickness mm (in) |
|---------------------|---------------------------|
| Bedding Material    | 150 (6)                   |
| General Fill        | 300 (12)                  |
| Structural Fill     | 200 (8)                   |



9. Compaction by water jetting or flooding shall not be permitted.
10. Operations shall be suspended when either the fill to be placed or the surface to receive fill is frozen, flooded or whenever the conditions of climate, materials, or equipment, as determined by the Contractors Geotechnical Engineer, are unsatisfactory for placing fill to the requirements of this GFS.
11. The testing shall be performed at frequencies indicated in Table 4.3.

**Table 4.3 Fill material test frequency and lift thickness**

| Material                                        | Frequency                                           | Lifts                    |
|-------------------------------------------------|-----------------------------------------------------|--------------------------|
| General fill                                    | 2 per 1000 m <sup>2</sup> (10,000 ft <sup>2</sup> ) | Minimum 1 per lift       |
| Structural fill                                 | 2 per 500 m <sup>2</sup> (5,000 ft <sup>2</sup> )   | Minimum 1 per lift       |
| Hand operated compaction                        | 1 per 80 m <sup>2</sup> (800 ft <sup>2</sup> )      | Minimum 1 per lift       |
| Confined areas (trenches and ground structures) | 1 per 80 m <sup>2</sup> (800 ft <sup>2</sup> )      | Minimum 1 per lift       |
| Fill under foundation                           | 1 per 80 m <sup>2</sup> (80 ft <sup>2</sup> )       | Minimum 1 per foundation |

12. At least one in-place moisture content and density test shall be performed on every lift of fill.
13. Further placement of fill shall not be allowed in case dry density has not been achieved.

#### 4.8 FINAL GRADING

1. The final grade shall be shaped to the required elevations and parallel to the required finished surface.
2. Line, level, elevations of excavations and final grading shall conform to that shown on the drawings within the tolerances stated in Table 4.4.
3. The closing error on all verification surveys and settlement monitoring surveys shall be less than +/- 3 mm (1/8 in) in elevation.
  - a. Where closure is greater than +/- 3 mm (1/8 in), the verification survey shall be repeated.
4. Upon completion of all installations including, structures, utilities, roads, paved areas, drainage and earthworks, the resulting surfaces shall be made uniformly smooth and graded.
5. Temporary construction facilities and lay down areas shall be graded to facilitate water removal towards perimeter drainage systems.

**Table 4.4 Dimensional tolerances in earthworks**

| <b>Description</b> | <b>Tolerance from true level in mm (in)</b> | <b>Tolerance from true line in mm (in)</b> |
|--------------------|---------------------------------------------|--------------------------------------------|
| Site clearance     | + 60 (2.3)<br>- 60 (2.3)                    | + 75 (3)<br>- 75 (3)                       |
| Excavations        | + 30 (1.2)<br>- 30 (1.2)                    | + 30 (1.2)<br>- 30 (1.2)                   |
| Rough grading      | + 5 (0.2)<br>- 25 (1)                       | + 75 (3)<br>- 75 (3)                       |
| Embankment         | + 75 (3)<br>- 0                             | + 75 (3)<br>- 75 (3)                       |

6. The Contractor shall make a smooth transition profile between areas where the original ground surface remains and areas which have been disturbed during construction.
7. The surface finishing of any slopes formed during construction shall conform with requirements shown on the drawings in Project SOW.

#### 4.9 DISPOSAL AND HAULAGE

1. All materials arising from demolition, site clearance works, clearing and grubbing and excavation except material specified for reuse, shall be removed from site as the work progresses and disposed of in accordance with the Project SOW.
2. Disposal of materials outside of the work premises shall be according to local regulations and project specification.
3. Contaminated soils and hazardous materials shall be managed in accordance with local regulations and project specification.
4. Burning of combustible material shall not be allowed on facility premises.
5. In-situ soil comprising of residual soils may be used as general fill material, however these in-situ soils shall not be used as structural fill.
6. Topsoil shall be reused in accordance with (4.10.2).

#### 4.10 SITE RESTORATION

##### 4.10.1 Cleaning

1. All areas affected by work shall be cleaned and reinstated to a condition, which existed prior to commencement of work.
2. The Contractor shall dispose of unsuitable and rejected materials in designated disposal areas only.
3. The Contractor shall rough grade disposed materials in the disposal area to ensure drainage and to give a tidy appearance.
4. The Contractor shall remove contaminated soil and other contaminants in accordance with the SOW.

**4.10.2 Topsoil Placement**

1. Topsoil placement scope shall consist of the excavation, loading, unloading and finishing of select topsoil material on slope ditched, medians or other areas identified in the Project SOW for establishing vegetation for erosion control.
2. All areas that are to be seeded or turfed shall be finished with topsoil.
3. The areas to be covered with topsoil shall be uniformly shaped and then scarified to a depth of 50 mm (2 in).
4. The required topsoil shall be taken from the topsoil stockpiles or imported and spread to a minimum depth of 100 mm (4 in).

**4.10.3 Seeding**

1. Seeding shall consist of preparing the area to be seeded, applying the seed and fertilizer, harrowing of the seeded area and protecting the seeded area by means of a mulch cover.
2. Areas to be seeded shall be fine graded to a smooth and uniform surface and to a loose depth of not less than 25 mm (1 in) at the time of seeding.
3. Seed, fertilizer and mulch shall be uniformly mixed and applied in accordance with the recommendations of an agronomist familiar with the local vegetation.
4. The surface preparation of the areas to be seeded shall be inspected to ensure compliance with recommendations of agronomist prior to commencement of seeding.
5. Within a designated seeded area; any area greater than 1 m<sup>2</sup> (10 ft<sup>2</sup>) that has less than 80 % vegetation ground cover shall be re-seeded by the Contractor.

**5. CONSTRUCTION MANAGEMENT****5.1 GENERAL**

1. The Contractor shall clearly specify within the Invitation to Tender and method statements, any inputs e.g. water requirements, laydown areas required for execution of the works.
2. The Contractor shall provide documents for review as contained in Table 6.1.

**5.2 HSSE AND SOCIAL PERFORMANCE**

1. The Contractor shall ensure that Contractor and sub-contractor personnel comply with the 12 Life Saving Rules and Shell golden rules, for all worksites.
2. The Contractor shall manage HSSE risks in line with the project HSSE management system and the HSSE policy.
  - a. The Contractor shall ensure implementation of the HSSE management system requirements.
  - b. HSSE activities undertaken by the Contractor as part of the work shall be fully documented.

- c. Prior to commencement of the work, the Contractor shall develop an HSSE Plan. As a minimum, this includes:
    - i. Risk Assessment including procedures and Job Hazard Assessment (JHA);
    - ii. monitoring and reporting requirements;
    - iii. incident management process;
    - iv. revisions and additions to the Contractor HSSE Plan;
    - v. incentive, improvements and consequence management;
    - vi. communication.
  - d. An Implementation and Training Plan shall be available prior to the start of work to ensure all the Contractor personnel, including sub-contractors, have the HSSE competencies necessary to perform the work.
  - e. The Contractor shall supply and maintain Personal Protective Equipment (PPE) for all staff and visitors to its construction sites.
    - i. PPE shall conform to local and international standards;
    - ii. as a minimum, all personnel shall wear steel-toed shoes or boots, coveralls, gloves, hard hats and safety glasses;
    - iii. specific PPE, such as a fall harness, face shield, chemical resistant gloves, hearing protection shall be available in addition to the basic PPE;
    - iv. Specific PPE shall be used when regulatory, HEMP or JHA indicates the requirement.
  - f. The Contractor shall provide medical and occupational health facilities at the worksites including the following:
    - i. adequate number of washbasins and toilets for staff;
    - ii. First Aid and medical care;
    - iii. rest shelter.
3. The Contractor shall take precautions to protect the environment and demonstrate/implement environmental controls to minimize the risk to the environment, in line with the environmental impact assessment and environmental permit conditions as per (2.2) including:
- a. Water Control Plan shall indicate the methods of controlling, removal and disposal of both surface runoff water and groundwater during the work.
  - b. Site Protection Plan shall include details for protecting the existing facilities and buildings and personnel against debris, dust and any other dangers during the work.
  - c. Environmental Management Plan shall outline the procedures and requirements to be implemented to minimize damages to the work site and include such details as water protection, erosion/sediment control, dust, pollution and contaminated spill control.
4. Where applicable, the Contractor shall ensure that all sub-contractors engaged in the delivery of the work will have a formal system of environmental protection in place which is at least equal to that of the Contractor plan.

### 5.3 MONTHLY REPORTING

1. The Contractor shall submit, as part of the Monthly Project Report, a detailed construction report that includes, at a minimum, the following:
  - a. detailed project schedule;
  - b. progress curve;
  - c. man-hour records;
  - d. activities performed during the previous month;
  - e. planned activities for the current month;
  - f. safety statistics for the previous month and project to date;
  - g. issues and concerns;
  - h. register showing status of engineering queries, deviation requests, site instructions and non-conformances;
  - i. register showing status of variation orders;
  - j. pictures.
2. The detailed project schedule shall outline the total days to complete the project and, at a minimum, include a schedule breakdown for the following (if specified as defined by the SOW):
  - a. preparation of engineering and procedures, including scheduled formal reviews, such as constructability reviews, HAZID/HAZOP and SIMOP reviews;
  - b. detailed milestones for draft and AFC installation procedures;
  - c. materials transport and receipt;
  - d. demobilisation.

### 5.4 DAILY PROGRESS REPORT

1. Daily progress reports shall be submitted for all activities.
2. The daily progress report shall include, at a minimum, the following information:
  - a. work completed over the last 24 hours;
  - b. planned work for the next 24 hours;
  - c. weather conditions including ambient air temperatures and precipitation;
  - d. progress;
  - e. HSSE summary report, including HSSE statistics;
  - f. QA/QC summary, including:
    - i. summary of the work inspected each day;
    - ii. substrate conditions under which the work was performed;
    - iii. summary of any nonconforming conditions detected.
  - g. concerns such as technical issues, schedule, materials;
  - h. personnel on site and man-hour records;
  - i. equipment status in the field.

## 5.5 QA/QC

1. QA/QC shall be the responsibility of the Contractor.
  - a. The Contractor shall have a Quality Management System (QMS) and Project Quality Plan (PQP).
  - b. A QMS shall consist of quality planning, control, assurance and continuous improvement processes.
  - c. The PQP shall fully describe how quality objectives will be managed for the duration of the work including those for the sub-contractors and/or the Manufacturers/Suppliers.
  - d. The Contractor shall provide the Principal access to all construction and quality records associated with the works.
2. The Contractor shall develop an Inspection and Test Plan(s) (ITP).
  - a. The ITP shall detail the inspection activities, reference to inspection procedures and/or acceptance criteria, and surveillance requirements for the Contractor, the Principal and independent inspection and testing agency.
4. During construction, the Contractor shall prepare written records of all inspections and tests and maintain a file of all records in a manner readily available to the job site.
  - a. These records shall contain ITPs with a list of the assigned personnel for the specific testing.
  - b. Respective QCRs for each completed activity shall be maintained in a file.
5. The Contractor shall handover a complete set of all documents to the Project in accordance with the Project requirements and not greater than within 30 days after completion of works.

### 5.5.1 Workmanship

1. The Contractor shall employ qualified and approved personnel, having experience in site preparation and earthworks.

### 5.5.2 Inspection and Approval

1. The Contractor shall employ an independent Inspection and Testing Agency to perform testing and soil evaluations to verify compliance of the work and workmanship with the requirements of the Project SOW and to ensure the achievement of the intents and purposes of the work.
2. The Inspection and Testing Agency shall, at a minimum, meet the requirements of ASTM D3740 and ASTM E329.
3. All inspection and testing shall be performed by or under the direct supervision of a registered (certified) Engineer or Technician experienced to perform the tasks.
4. The Inspection and Testing Agency shall monitor all earthwork operations to ensure the work is accomplished in accordance with all Project SOW.
5. The inspection and testing agency shall record all test results in a spreadsheet and shall provide both statistical assessment and graphical trends of the results by each material type.
  - a. The results shall be included in both weekly and monthly reports.
6. Any nonconforming conditions detected, which are not immediately brought into conformance by the Contractor, shall be documented in a written report together with proposed mitigations.

**5.6 MEETINGS**

1. During construction, the Contractor shall hold daily and weekly production meetings with the Principal to discuss, as a minimum, the following subjects:
  - a. HSSE (previous day's report, concerns, opportunities);
  - b. previous day's activities;
  - c. planned activities for the next day and look ahead for the next 10 days;
  - d. actual production rates and accumulated production to date vs. planned production rate (i.e., impact on schedule);
  - e. concerns;
  - f. written meeting minutes shall be issued along with the status of action items reviewed during the daily production meeting.
2. Before any work starts, a toolbox talk shall be held, explaining the operations in a structured way emphasizing the safety measures to be taken to execute the work safely.
  - a. All personnel executing the job shall sign the toolbox talk attendance sheet.
3. Contractor shall participate in any required meetings with the Principal necessary to plan works.

**5.7 PERMIT TO WORK**

1. A Permit to Work (PTW) system shall be used for all construction activities carried out by the Contractor as part of the work.
2. Where applicable Contractor shall train and develop staff to comply with the requirements of the Project PTW system.

**5.8 OFFICE FACILITIES**

1. Contractor shall provide offices equipped with phone, internet and copy facilities readily available for as per Project requirements.

**6. DOCUMENTATION****6.1 DOCUMENT PURPOSE**

*This section provides the minimum deliverable requirements for site preparation and earthworks. Table 6.1 specifies whether a deliverable is submitted for review or acceptance.*

1. Documentation shall be submitted for review in conformance to Table 6.1.

**Table 6.1 Deliverable requirements**

| <b>Deliverable Type</b>                             | <b>Deliverable Title</b> | <b>Principal Action</b> | <b>Comments</b>                                                                                 |
|-----------------------------------------------------|--------------------------|-------------------------|-------------------------------------------------------------------------------------------------|
| <b>Documents to be Submitted with Technical Bid</b> |                          |                         |                                                                                                 |
| Schedule                                            | Level 3 Schedule         | Review                  | To be included in Technical Bid                                                                 |
| Specification                                       | Imported Fill Material   | Accept                  | Include in Technical Bid:<br>Source and Material Specifications to be provided in technical Bid |

| Equipment                                            | List of Equipment to be utilised for work                                                                                                                                                                                                                                      | Review | To be included in Technical Bid |
|------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|---------------------------------|
| <b>Documents to be Submitted Before Mobilisation</b> |                                                                                                                                                                                                                                                                                |        |                                 |
| Project Plan                                         | Project Execution Plan                                                                                                                                                                                                                                                         | Accept |                                 |
|                                                      | Organization, responsibilities and supervision                                                                                                                                                                                                                                 | Accept |                                 |
|                                                      | Material Plan<br>i. Information on the proposed source of fill material.<br>ii. Properties of fill materials including sieve analysis data.<br>iii. Information of proposed site of borrow materials and the properties of the borrow materials including sieve analysis data. | Accept | 3                               |
|                                                      | i. Information on the proposed source of water.<br>ii. Properties of water.                                                                                                                                                                                                    | Accept | 3                               |
|                                                      | Mix Design for CLSM                                                                                                                                                                                                                                                            | Accept | 3.5                             |
|                                                      | Stockpiling plan (if applicable)                                                                                                                                                                                                                                               | Accept | 4.4.2                           |

|          |                                                                                                                                                 |        |       |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------------|--------|-------|
|          | HSSE Plan<br>(including environmental management plan, water control plan, waste management plan including disposal)<br>Emergency Response Plan | Accept |       |
|          | Mobilisation Plan                                                                                                                               | Accept |       |
|          | Site Protection Plan (including risk assessment for personnel and existing facilities protection).                                              | Accept |       |
|          | Project Quality Plan (including Inspection and Test Plans)                                                                                      | Accept |       |
|          | Change Control Procedure                                                                                                                        | Accept |       |
|          | Method statements and equipment for work including Equipment Sparing Philosophy                                                                 | Accept |       |
|          | Erosion Control Plan                                                                                                                            | Accept | 4.1.4 |
|          | Drainage Plan                                                                                                                                   | Accept | 4.1.5 |
| Register | Master Document Register                                                                                                                        | Review |       |



| Documents to be Submitted During Works                  |                                               |        |     |
|---------------------------------------------------------|-----------------------------------------------|--------|-----|
| Report                                                  | Calibration Report on Trial Area              | Accept | 4.5 |
| Report                                                  | Monthly Project Report (5.3)                  | Review | 5.3 |
| Report                                                  | Daily Progress Report (5.4)                   | Review | 5.4 |
| Records                                                 | Inspection Test Records (5.5)                 | Review | 5.5 |
| Documents to be Submitted following Completion of Works |                                               |        |     |
| Records                                                 | Complete Set of Inspection Test Records (5.5) | Review | 5.5 |
| Drawings                                                | As Built Drawings                             | Review |     |

**7. REFERENCES**

In this GFS, reference is made to the following publications:

- NOTES:
1. Unless specifically designated by date, the latest edition of each publication shall be used, together with any amendments/supplements/revisions thereto.
  2. The SPEPs, GFSs and most referenced external standards are available to Shell staff on the SWW (Shell Wide Web) at <http://sww.shell.com/standards/>.

**AMERICAN STANDARDS**

|                                                                                                                                                              |                                            |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------|
| Report on Controlled Low-Strength Materials                                                                                                                  | American Concrete Institute (ACI) ACI 229R |
| Standard Test Method for Sieve Analysis of Fine and Coarse Aggregates                                                                                        | ASTM C136                                  |
| Test Method for Measurement of Solids in Water                                                                                                               | ASTM C1603                                 |
| Standard Test Method for Resistance to Degradation of Large-Size Coarse Aggregate by Abrasion and Impact in the Los Angeles Machine                          | ASTM C535                                  |
| Standard Test Method for Soundness of Aggregates by Use of Sodium Sulphate or Magnesium Sulphate                                                             | ASTM C88                                   |
| Standard Test Method for Density and Unit Weight of Soil in Place by Sand-Cone Method                                                                        | ASTM D1556                                 |
| Standard Test Methods for Laboratory Compaction Characteristics of Soil Using Modified Effort                                                                | ASTM D1557                                 |
| Standard Test Method for Density and Unit Weight of Soil in Place by the Rubber Balloon Method                                                               | ASTM D2167                                 |
| Standard Test Methods for Laboratory Determination of Water (Moisture) Content of Soil and Rock by Mass                                                      | ASTM D2216                                 |
| Standard Practice for Classification of Soils for Engineering Purposes (Unified Soil Classification System)                                                  | ASTM D2487                                 |
| Standard practice for description and identification of soils (visual-manual procedure)                                                                      | ASTM D2488                                 |
| Standard Test Method for Density of Soil in Place by the Drive-Cylinder Method                                                                               | ASTM D2937                                 |
| Standard Practice for Minimum Requirements for Agencies Engaged in Testing and/or Inspection of Soil and Rock as Used in Engineering Design and Construction | ASTM D3740                                 |
| Standard Test Methods for Maximum Index Density and Unit Weight of Soils Using a Vibratory Table                                                             | ASTM D4253                                 |
| Standard Test Methods for Minimum Index Density and Unit Weight of Soils and Calculation of Relative Density                                                 | ASTM D4254                                 |
| Standard Test Methods for Liquid Limit, Plastic Limit, and Plasticity Index of Soils                                                                         | ASTM D4318                                 |
| Standard Test Methods for Density of Soil and Rock in Place by the Sand Replacement Method in a Test Pit                                                     | ASTM D4914                                 |
| Standard Test Methods for Density of Soil and Rock in Place by the Water Replacement Method in a Test Pit                                                    | ASTM D5030                                 |
| Standard Test Methods for Chloride Ion in Water                                                                                                              | ASTM D512                                  |

|                                                                                                                            |            |
|----------------------------------------------------------------------------------------------------------------------------|------------|
| Standard Test Method for Sulphate Ion in Water                                                                             | ASTM D516  |
| Standard Test Method for Particle-Size Analysis of Soils                                                                   | ASTM D6913 |
| Standard Test Methods for In-Place Density and Water Content of Soil and Soil-Aggregate by Nuclear Methods (Shallow Depth) | ASTM D6938 |
| Standard Test Methods for Laboratory Compaction Characteristics of Soil Using Standard Effort                              | ASTM D698  |
| Standard Specification for Agencies Engaged in Construction Inspection, Testing, or Special Inspection                     | ASTM E329  |
| Standard Test Method for Measuring pH of Soil for Use in Corrosion Testing                                                 | ASTM G51   |

### PROJECT STANDARDS

The project and Engineering Contractor shall apply the standards that have been specified above and no project-specific technical standards and/or specifications shall be made on top of these.

*It is recognized that the Engineering Contractor has its own procedures and standards to manage a project, from that perspective it is allowed to develop a limited number of project specific standards, specifications, procedures that are required to manage and administer the project execution.*